

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTHM505 & Project Management

Max Marks: 60

Date: 10/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level /CO)	Marks																									
Q. 1 Solve Any Two of the following.			12																									
A) Explain in detail project, project management and components of project management.		CO1	6																									
B) Explain the uses, advantages, and limitations of break-even analysis		CO5	6																									
C) Discuss the importance of quality control in Project Management		CO5	6																									
Q.2 Solve Any Two of the following.			12																									
A) Explain Resource Allocation		CO2	6																									
B) The following table gives the data for the duration and costs of each activity of project network. The indirect cost of the project 2000Rs/Week.		CO2	6																									
<table><tr><th>Activity</th><th>Normal Duration (Week)</th><th>Normal Cost (Rs.)</th><th>Crash Duration (Week)</th><th>Crash Cost (Rs.)</th></tr><tr><td>1-2</td><td>4</td><td>4000</td><td>2</td><td>12000</td></tr><tr><td>2-3</td><td>5</td><td>3000</td><td>2</td><td>7500</td></tr><tr><td>2-4</td><td>7</td><td>3600</td><td>5</td><td>6000</td></tr><tr><td>3-4</td><td>4</td><td>5000</td><td>2</td><td>10000</td></tr></table>				Activity	Normal Duration (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Crash Cost (Rs.)	1-2	4	4000	2	12000	2-3	5	3000	2	7500	2-4	7	3600	5	6000	3-4	4	5000	2	10000
Activity	Normal Duration (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Crash Cost (Rs.)																								
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2-4	7	3600	5	6000																								
3-4	4	5000	2	10000																								
Draw network. Find project duration and show critical path based on normal duration and corresponding total project cost. Calculate cost slope for each activity.																												
C) Determine the optimum duration of project and the corresponding minimum cost associated with the project in Q.no.2 (B)		CO2	6																									
Q. 3 Solve Any Two of the following.			12																									
A) Define and explain Optimistic time, Pessimistic time, and Most likely time.		CO3	6																									

B) Draw the network and find the project duration & Critical path for the following

CO3

6

Activity (i-j)	t_o (Optimistic Time)	t_L (Most likely Time)	t_p (Pessimistic Time)
1-2	2	5	8
2-3	8	11	20
3-4	0	0	0
2-4	4	7	16
2-5	4	9	20
4-6	7	10	13
5-6	3	7	17
3-7	3	5	13
6-7	2	3	10
7-8	2	4	6

C) Find Standard Deviation and Z value if the project mentioned in Q. no 3 (B) has the completed in 40 Days.

CO3

6

Q.4 Solve Any Two of the following.

12

A) Explain Demand and Supply

CO4

6

B) Explain Inflation and Cause of Inflation.

CO4

6

C) Explain Time value of money.

CO4

6

Q. 5 Solve Any Two of the following.

12

A) Draw the network and calculate TE & TL for all activities for the following.

CO1

6

Activity (i-j)	1-2	1-3	1-4	2-5	2-7	3-4	3-6	4-5	5-6	5-7	6-7	7-8
Duration (t^{ij}) Weeks	6	7	8	7	8	4	4	9	5	6	6	9

B) Prepare table and calculate EST, EFT, LST, LFT, and Total Float associated with the project in Q.no.5(A), Find critical path & project Duration

CO1

6

C) Explain break even analysis.

CO5

6

***** End *****